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/*
  SparkFun Inventor's Kit
  Circuit 4B - Temperature Sensor

  The LCD will display readings from a temperature sensor in degrees Celsius
and Fahrenheit.

  This sketch was written by SparkFun Electronics, with lots of help from the
Arduino community.
  This code is completely free for any use.

  View circuit diagram and instructions at: https://learn.sparkfun.
com/tutorials/sparkfun-inventors-kit-experiment-guide---v41
  Download drawings and code at: https://github.com/sparkfun/SIK-Guide-Code
*/

#include <LiquidCrystal.h>                    //the liquid crystal library
contains commands for printing to the display
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);    // tell the RedBoard what pins are
connected to the display

float voltage = 0;                            //the voltage measured from the
TMP36
float degreesC = 0;                           //the temperature in Celsius,
calculated from the voltage
float degreesF = 0;                           //the temperature in Fahrenheit,
calculated from the voltage

void setup() {

  lcd.begin(16, 2);                           //tell the lcd library that we
are using a display that is 16 characters wide and 2 characters high
  lcd.clear();                                //clear the display
}

void loop() {

  voltage = analogRead(A0) * 0.004882813;    //convert the analog reading,
which varies from 0 to 1023, back to a voltage value from 0-5 volts
  degreesC = (voltage - 0.5) * 100.0;         //convert the voltage to a
temperature in degrees Celsius
  degreesF = degreesC * (9.0 / 5.0) + 32.0; //convert the voltage to a
temperature in degrees Fahrenheit
```

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  lcd.clear();                              //clear the LCD

  lcd.setCursor(0, 0);                      //set the cursor to the top left
position
  lcd.print("Degrees C: ");                 //print a label for the data
  lcd.print(degreesC);                      //print the degrees Celsius

  lcd.setCursor(0, 1);                      //set the cursor to the lower
left position
  lcd.print("Degrees F: ");                 //Print a label for the data
  lcd.print(degreesF);                      //print the degrees Fahrenheit

  delay(1000);                              //delay for 1 second between each
reading (this makes the display less noisy)
}
```